

Mitchell C. Elliott

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Software Engineer | Santa Cruz, CA 95060

SUMMARY

Software engineer specializing in building scalable and sustainable systems across storage, AI, and cloud infrastructure. Experienced in R&D, DevOps, and full-stack development, focusing on performance, efficiency, and real-world impact.

SKILLS

Languages: Python, C, C++, Java, JavaScript, Bash, SQL

Infrastructure & Platforms: Linux, Docker, Kubernetes, AWS, Terraform, Jenkins, Kafka, Grafana, New Relic, Git, Jira

Databases & Storage: PostgreSQL, MySQL, Redis, OpenSearch, SQLite, MongoDB, Cassandra, MinIO, Ceph, DuckDB

Frameworks & Libraries: React, Node.js, Flask, FastAPI, PyTorch, Pandas

EXPERIENCE

University of California, Santa Cruz

Jan. 2026 – Present

Research and Development Engineer

Santa Cruz, CA

- Leading a 3-person engineering team building modeling and analysis tools for sustainable data-center storage systems
- Designing hybrid storage architectures optimizing carbon, cost, and performance across production-scale data-center workloads
- Developing simulation platforms and telemetry data pipelines for large-scale evaluation of storage-system architectures

University of California, Santa Cruz

Dec. 2024 – Jan. 2026

Postgraduate Researcher

Santa Cruz, CA

- Designed and evaluated sustainable storage-system architectures balancing carbon, cost, and performance tradeoffs
- Applied life cycle assessment (LCA) frameworks to model energy and environmental impact across storage technologies
- Built simulation and systems-analysis tooling for workload-aware evaluation of storage configurations and design tradeoffs

University of California, Santa Cruz

Jan. 2023 – Dec. 2024

Teaching Assistant

Santa Cruz, CA

- Served as Head TA for CSE 130: Principles of Computer System Design across 6 quarters, mentoring 1000+ students
- Led discussion sections, developed assignments and quizzes, proctored exams, and managed course operations
- Built and maintained automated GitLab CI/CD grading infrastructure for programming assignments, syncing scores with Canvas

Perfectly Snug

July 2023 – Sept. 2023

Software Engineer Intern

Remote

- Built internal business systems for material planning and BOM management, supporting thousands of customer orders
- Automated inventory tracking and manufacturing workflows, improving operational efficiency and reducing production delays

ParkourSC, Inc.

July 2022 – Sept. 2022

DevOps Engineer Intern

Remote

- Redesigned alerting policies across dozens of Kubernetes clusters in New Relic, improving observability and signal quality
- Reduced false positives in infrastructure monitoring, lowering incident noise and improving focus on critical production issues

uLab Systems, Inc.

June 2021 – Sept. 2021

Software Engineer Intern

San Mateo, CA

- Migrated WordPress infrastructure from Amazon Lightsail to EC2, improving deployment flexibility and maintainability
- Designed and implemented a three-stage CI/CD pipeline, increasing release reliability across a multi-service cloud environment

PROJECTS

Workload Trace Analysis Platform

Jan. 2026 – Present

- Built a Kubernetes-based analytics platform for querying and analyzing multi-terabyte Google workload trace datasets
- Created reusable tooling for concurrent multi-user evaluation, reducing cloud costs by hundreds to thousands of dollars

Data Center Carbon TCO Simulator

Jan. 2024 – Present

- Built a Python-based simulator that models end-to-end total cost of ownership (TCO) for data centers
- Developed a multi-metric optimization framework integrating cost, performance, and sustainability to guide tiered storage layouts

EDUCATION

University of California, Santa Cruz

M.S. Computer Science | Sept. 2022 – Dec. 2024

B.S. Computer Science | Sept. 2018 – June 2022